

Overview of Genetics



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Course Outline

- ☐ Overview of Genetics.
- ☐ A Darwinian view of life.
- ☐ Levels of selection.
- ☐ Genetic Variation.
- ☐ Population genetics.
- ☐ Evidence of Evolution.
- ☐ Introduction to molecular cloning.
- ☐ Molecular Cloning Methods.
- ☐ Gene Therapy.

Learning Outcomes (LOs)

- ☐ Overview of Genetics.
- ☐ Genetic technologies allow modification of traits of animals & plants.
- ☐ Molecular organization of a living cell.
- ☐ The genetic material in living organisms

Overview of Genetics

- ☐ **Genetics** is the study of heredity and variation
- ☐ **Genetics** is the unifying discipline in biology – leads to understanding how life can exist at all levels of complexity
- ☐ Explains why organisms have certain **traits**
 - How **traits** are passed from parent to offspring
 - Relationship between **genes** and **traits**

Genetic Discoveries and Technology

- ❑ Genetic discoveries are growing at a fast pace.
- ❑ Example: **The Human Genome Project**.
 - Goal: interpret the information in the **human genome** – the DNA of all our chromosomes.
 - Published in 2003.
 - 3 billion **nucleotide** base pairs.
 - Shed light on basic questions about **human genes**, traits, and possible cures for disease.

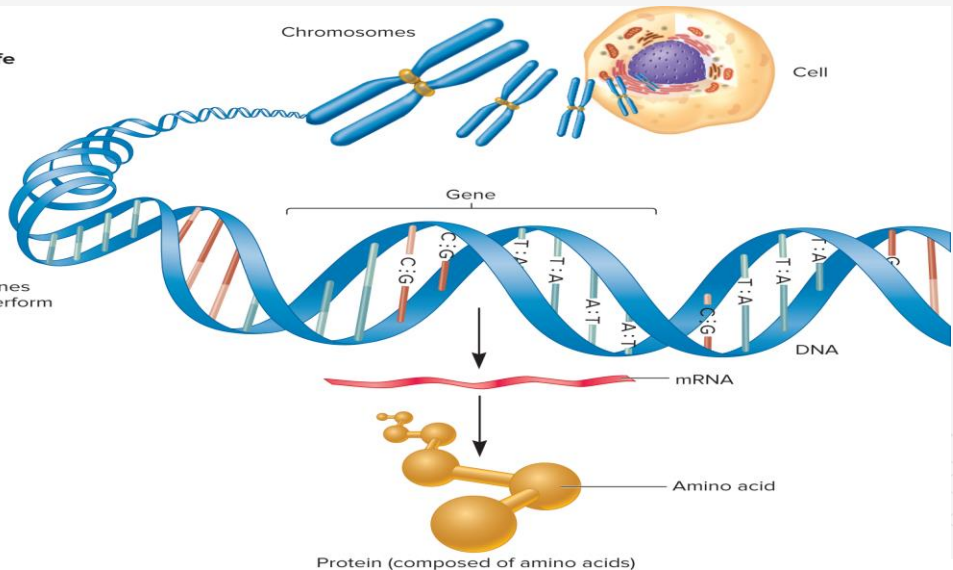
The Human Genome

DNA, the molecule of life

The adult human body is composed of trillions of cells.

Most human cells contain the following:

- 46 human chromosomes, found in 23 pairs
- 2 meters of DNA
- Approximately 22,000 genes coding for proteins that perform most life functions
- Approximately 3 billion DNA base pairs per set of chromosomes, containing the bases A, T, G, and C



Genetic Technology

❑ New genetic technologies are useful and sometimes controversial.

❑ Mammalian cloning:

- 1997, Dolly the sheep was cloned.
- 2002, Carbon copy (or “Copycat”).
- Cloned livestock could benefit farmers.
- Human cloning has been legally banned.

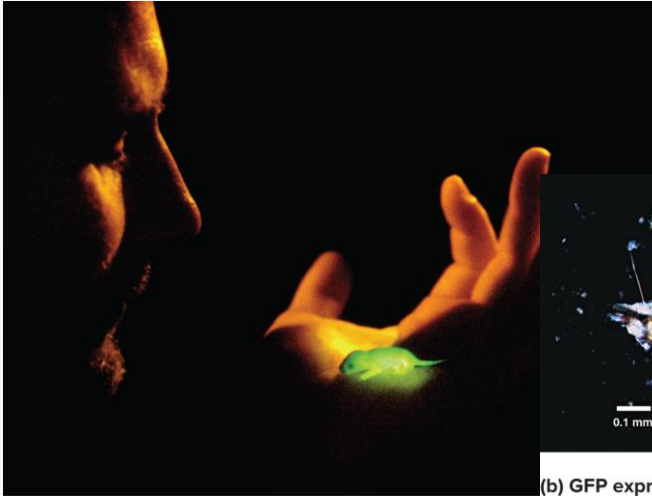


Genetic technologies allow modification of traits of animals & plants

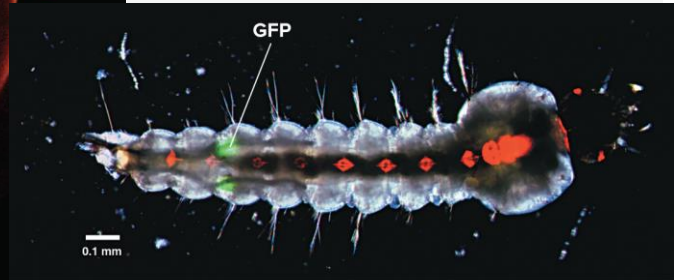
❑ Green fluorescent protein (GFP)

- A **jellyfish** gene produces a **bioluminescent protein**.
- Under UV light it emits a **green glow**
- 2008 Nobel Prize was awarded for the development of this powerful tool
- Examples:
 - **GFP** gene expressed in **mice**.
 - **GFP** expressed in the **gonads of male mosquitoes**.
 - Could be used to select sterile males for malaria control programs.

GFP gene expression



(a) GFP expressed in mice



(b) GFP expressed in the gonads of a male mosquito

The Molecular Expression of Genes

- ☐ Biochemical composition of cells.
- ☐ How DNA stores the information to make proteins?
- ☐ Proteins are largely responsible for cells structure and function.

Genes and Traits

- ❑ Geneticists study the relationship between **genes** and **traits**.
- ❑ **Gene:** a segment of **DNA** that has the information to produce a **functional protein**.
- ❑ **Trait:** any **characteristic** that an organism displays.
- ❑ During growth and development, **genes** provide a blueprint that determine the organism's **traits**.
- ❑ **Examples of human traits:**
 - Eye color.
 - Hair texture.
 - Height.

Living Cells are Composed of Biochemicals

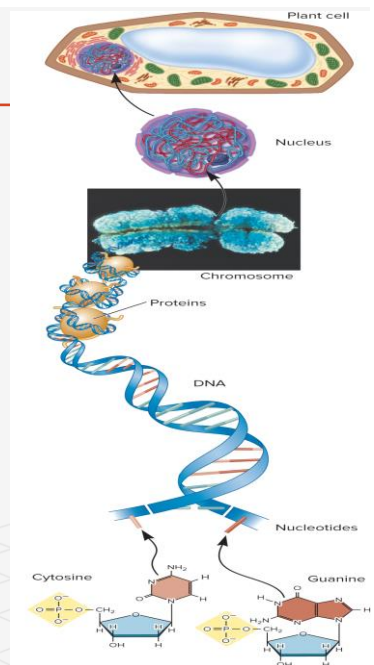
- ❑ All cells are constructed from **chemical substances**.
- ❑ **Organic molecules** are linked together by **chemical bonds** to form **larger molecules**.
- ❑ **Cells contain four main types of large molecules:**
 - Nucleic acids
 - Proteins
 - Carbohydrates
 - Lipids

Living Cells are Composed of Biochemicals

- ❑ Nucleic acids, proteins and carbohydrates can form **macromolecules**.
 - Many repeating units of smaller building blocks.
- ❑ Cellular structures form as a result of the interaction of **molecules** and **macromolecules**.
- ❑ **Organelle**: a membrane-bound compartment with a specialized function.
 - example:
 - nucleus**: protects chromosomes and compartmentalizes genetic activities

Molecular organization of a living cell

- ❑ **Nucleotide** building blocks link together to form long strands of **DNA**.
- ❑ **DNA** associates with **proteins** to form **chromosomes**.
- ❑ **Chromosomes** are contained within the nucleus of eukaryotic cells.



Proteins Determine Cell Structure and Function

- ❑ The characteristics of a cell depend on its **proteins** - “workhorses” of cells.
- ❑ **Proteome**: collection of **proteins** that a cell makes at a given time.
- ❑ They have diverse **biological functions**:
 - Cell shape and structure.
 - Transport.
 - Motor proteins
 - Signaling.
 - Enzymes: proteins that speed up chemical reactions.

DNA Stores the Information for Protein Synthesis

- ❑ The genetic material in living organisms is **deoxyribonucleic acid (DNA)**
- ❑ **DNA** stores the information required to synthesize all cellular **proteins**
 - It is able to do so because of its molecular structure.
 - **DNA** is a polymer of **nucleotides**.

Genetic code

- ☐ Each **nucleotide** contains one **nitrogenous base**:
 - Adenine (A)
 - Thymine (T)
 - Cytosine (C)
 - Guanine (G)
- ☐ Genetic information is stored in the linear sequence of **bases**.
- ☐ **DNA** sequence within **genes** contain information to direct the order of **amino acids** within **polypeptides** according to the **genetic code**.

Genetic code

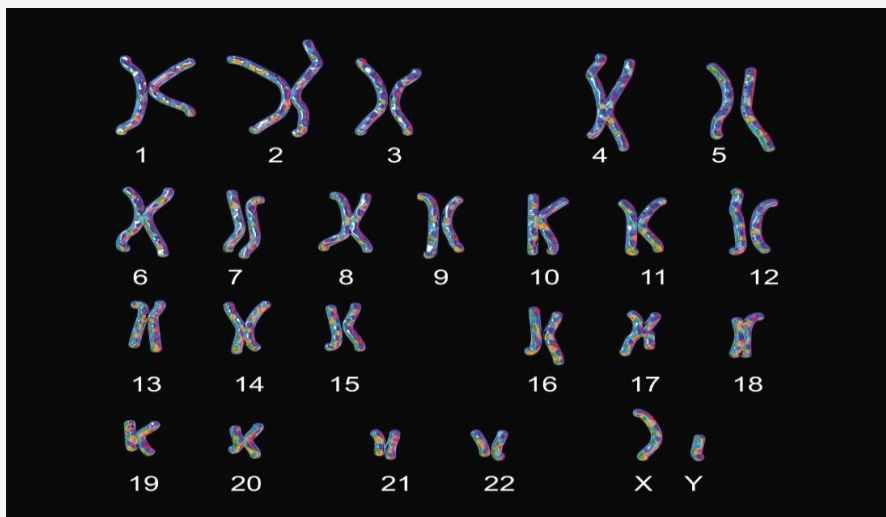
- ☐ In the **genetic code**, a three-base sequence, called a **codon**, specifies one particular **amino acid**.
- ☐ Sequence of **amino acids** causes a **polypeptide** to fold into a particular structure.
- ☐ For example:

DNA Sequence	Amino Acid Sequence
ATG GGC CTT AGC	Methionine Glycine Leucine Serine
TTT AAG CTT GCC	Phenylalanine Lysine Leucine Alanine

Chromosomes

- ❑ The **DNA** in living cells is contained within large structures termed **chromosomes**.
- ❑ Each **chromosome** is a complex of **DNA** and **proteins**.
- ❑ An average human **chromosome** contains:
 - More than a 100 million **nucleotides**.
 - about 1,000 different **genes**.
- ❑ Human cells have a total of **46 chromosomes**, found in **pairs**.

Chromosomes



DNA Information is Accessed During the Process of Gene Expression

- ❑ **Gene expression** is the use of a **gene** sequence to affect the **characteristics** of cells and organisms.
- ❑ **Occurs in steps:**
 - **Transcription:** The genetic information in DNA is copied into a nucleotide sequence of ribonucleic acid (RNA).
 - **Translation:** The nucleotide sequence in RNA provides the information (using the genetic code) to produce the amino acid sequence of a polypeptide.
 - **Polypeptide** folds into 3D structure.
 - Functioning of **proteins** determines cell structure & function.

DNA Information is Accessed During the Process of Gene Expression

